Z + jets Run 3: before and after, all campaigns

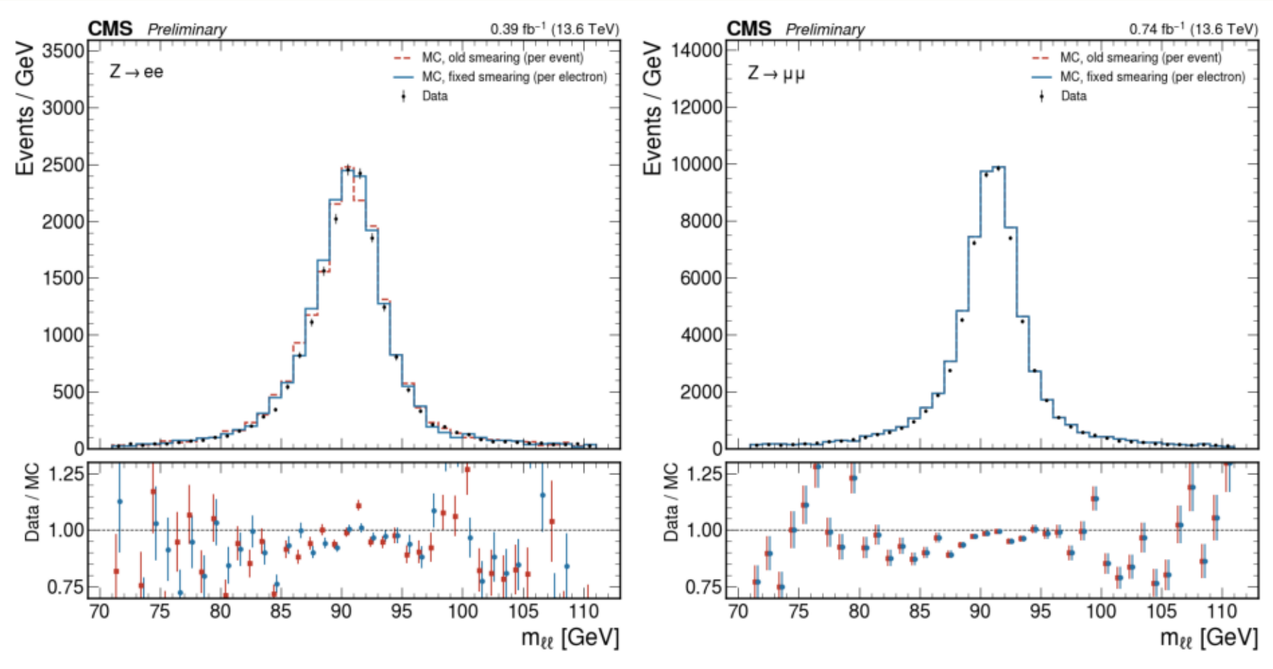
Z mass (electron smearing fix) and leading-lepton pT (MET fix + Z pT weights)

Small run: 20 files per sample · 2022 C–D, 2022 E–G, 2023 C, 2023 D

29 October 2026

Z mass, 2022 C–D: electron smearing before and after

Red dashed: MC with the old smearing. Blue: MC with the fix. Muons are not touched by this fix.



	before	after
Z→ee width, data/MC	0.95	0.99
Z→ee peak, data − MC (GeV)	+0.05	+0.05
Z→μμ peak, data − MC (GeV)	+0.04	+0.04

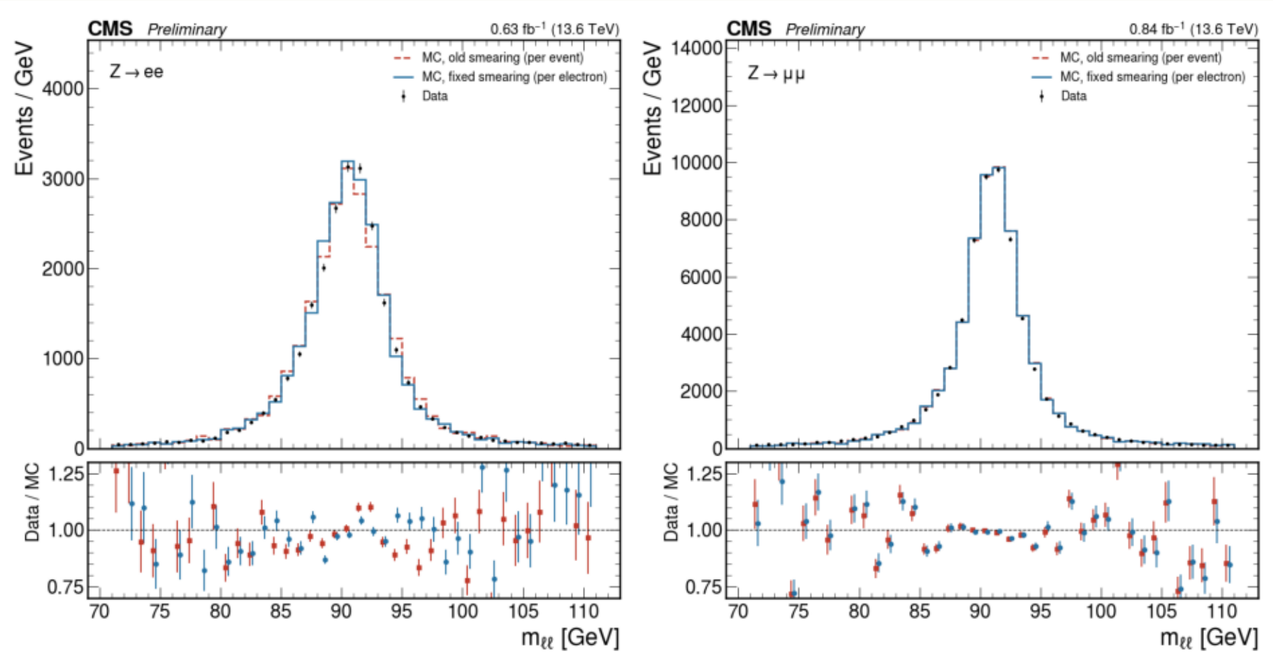
Width: 0.95 → 0.99. Peak position: unchanged.

The fix only changes the width. The electron peak offset (< 0.1%) is inside the EGM energy-scale uncertainty (0.05–0.36%).

Run 3, 13.6 TeV · Z→ℓℓ + jets · 20 files per MC sample and per data channel · AK4 jets · 2

Z mass, 2022 E–G: electron smearing before and after

Red dashed: MC with the old smearing. Blue: MC with the fix. Muons are not touched by this fix.



	before	after
Z→ee width, data/MC	0.94	1.00
Z→ee peak, data − MC (GeV)	+0.06	+0.07
Z→μμ peak, data − MC (GeV)	−0.03	−0.03

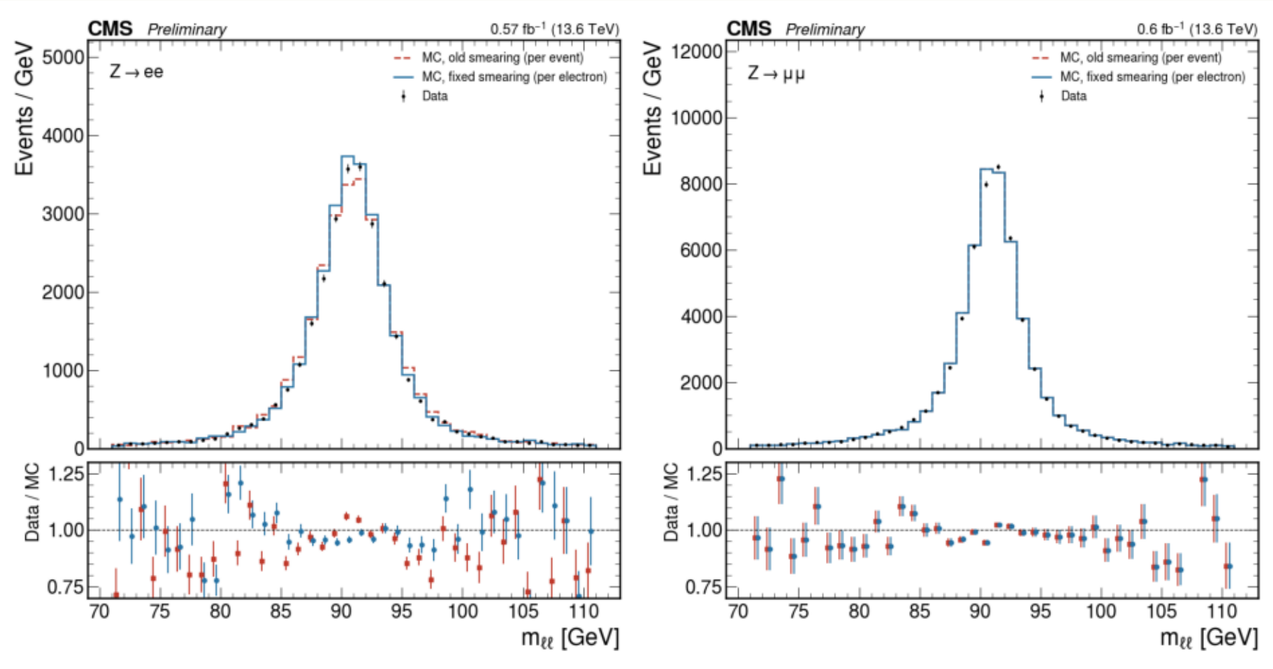
Width: 0.94 → 1.00. Peak position: unchanged.

The fix only changes the width. The electron peak offset (< 0.1%) is inside the EGM energy-scale uncertainty (0.05–0.36%).

Run 3, 13.6 TeV · Z→ℓℓ + jets · 20 files per MC sample and per data channel · AK4 jets · 3

Z mass, 2023 C: electron smearing before and after

Red dashed: MC with the old smearing. Blue: MC with the fix. Muons are not touched by this fix.



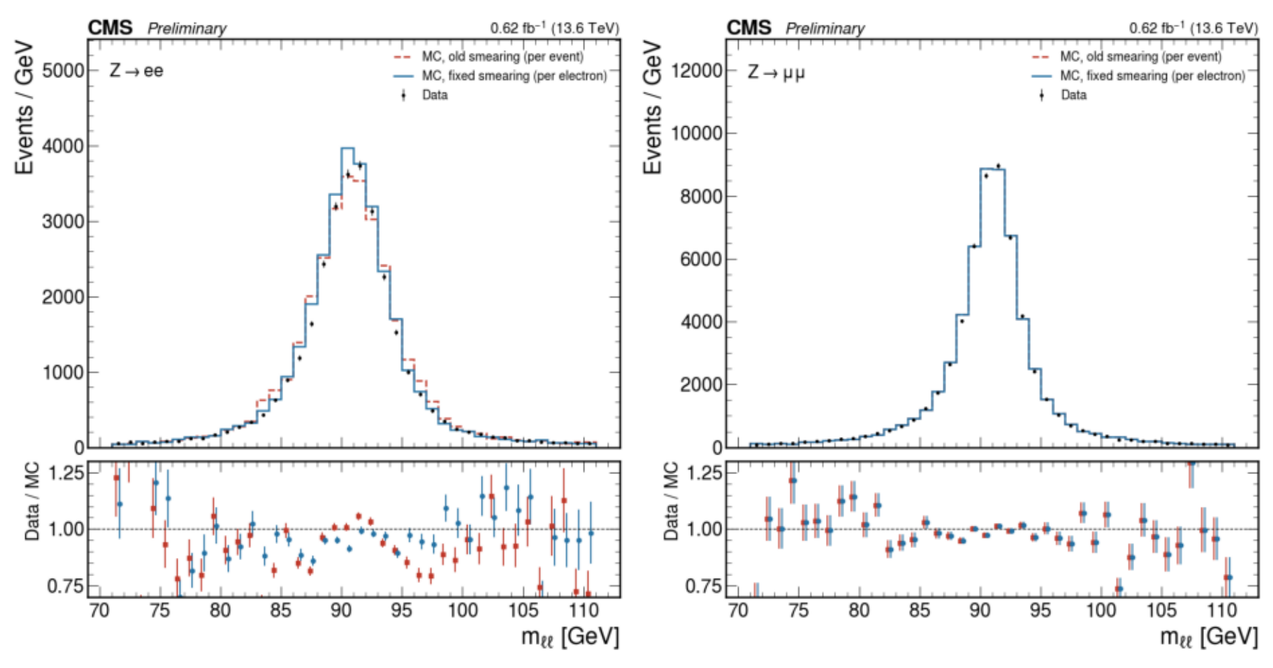
	before	after
Z→ee width, data/MC	0.94	1.00
Z→ee peak, data − MC (GeV)	+0.02	+0.03
Z→μμ peak, data − MC (GeV)	+0.03	+0.03

Width: 0.94 → 1.00. Peak position: unchanged.

The fix only changes the width. The electron peak offset (< 0.1%) is inside the EGM energy-scale uncertainty (0.05–0.36%).

Z mass, 2023 D: electron smearing before and after

Red dashed: MC with the old smearing. Blue: MC with the fix. Muons are not touched by this fix.



	before	after
Z→ee width, data/MC	0.90	0.97
Z→ee peak, data − MC (GeV)	+0.08	+0.08
Z→μμ peak, data − MC (GeV)	+0.03	+0.03

Width: 0.90 → 0.97. Peak position: unchanged.

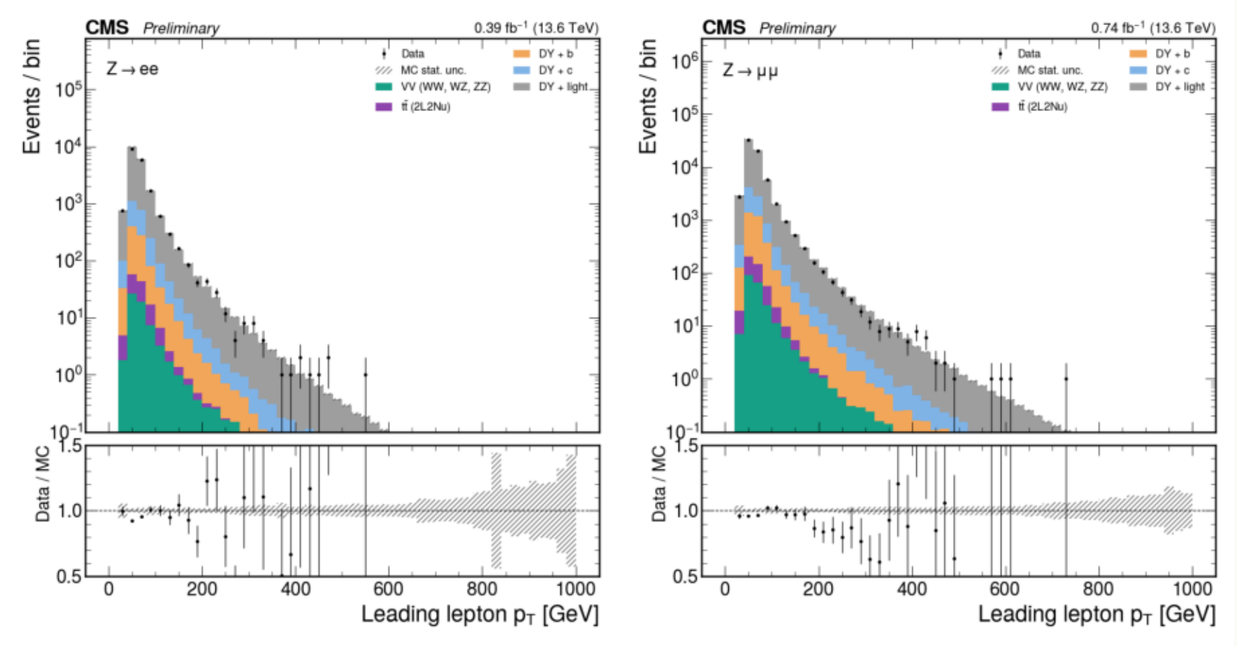
The fix only changes the width. The electron peak offset (< 0.1%) is inside the EGM energy-scale uncertainty (0.05–0.36%).

Run 3, 13.6 TeV · Z→ℓℓ + jets · 20 files per MC sample and per data channel · AK4 jets · 5

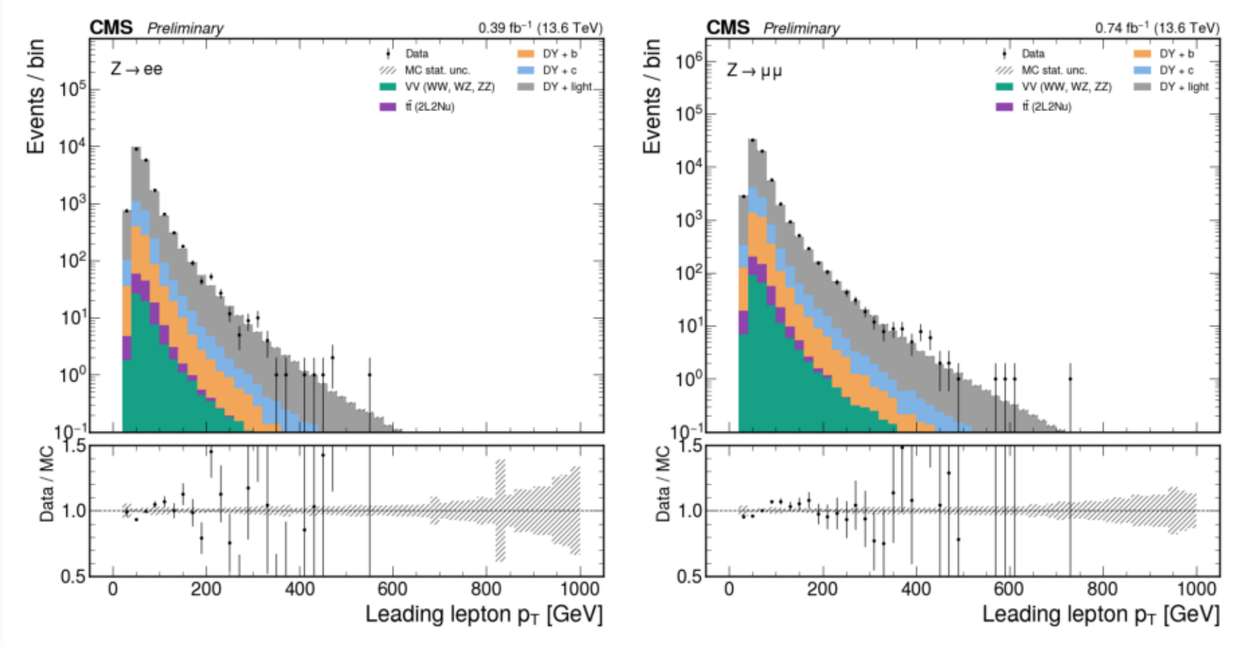
Leading-lepton pT, 2022 C–D: before and after

Left: before. Right: after (electrons' jets out of the MET + DY Z pT weights). Same files.

Before



After



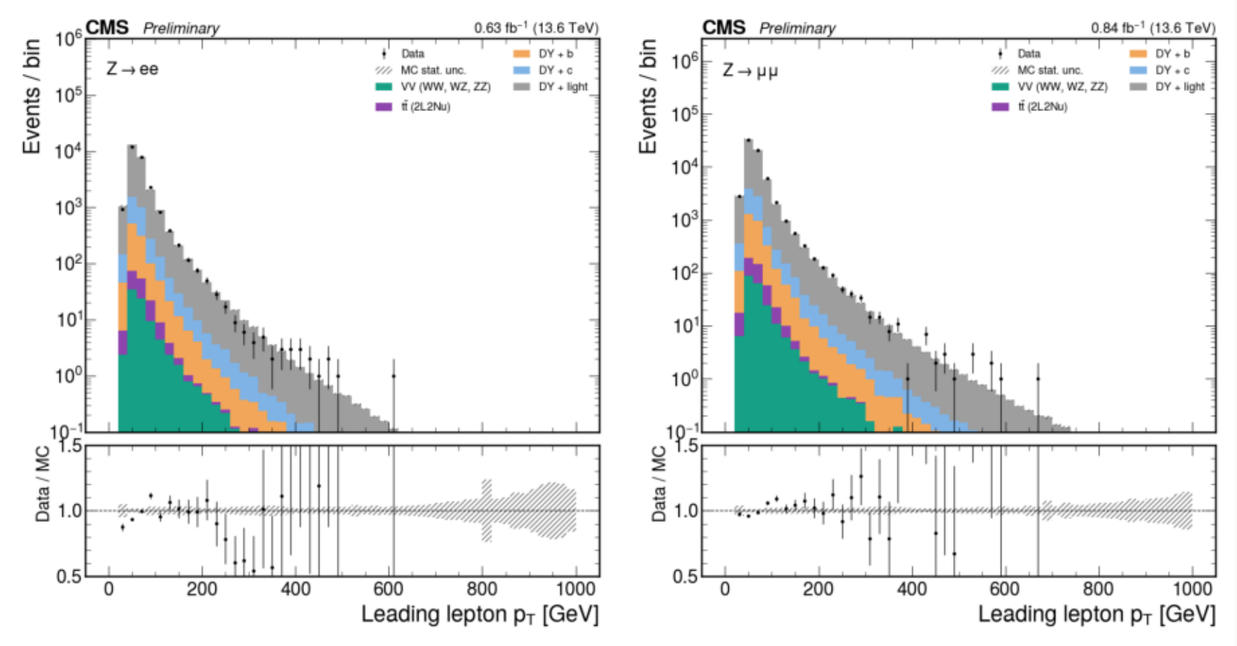
Level about 2% higher; shape not better

Z→ee 60–200 GeV: 0.96–0.98 → 1.01–1.05. Z→μμ 200–300 GeV: 0.83 → 0.97. Below 60 GeV still 0.94–0.96 in both channels.

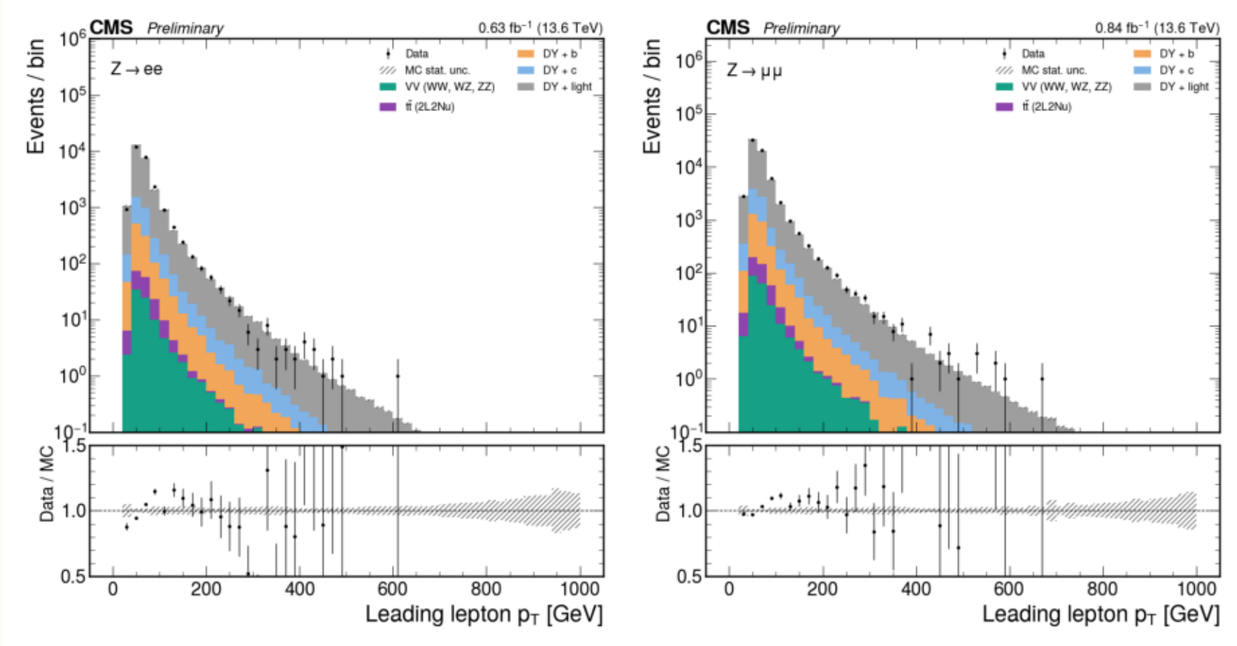
Leading-lepton pT, 2022 E–G: before and after

Left: before. Right: after (electrons' jets out of the MET + DY Z pT weights). Same files.

Before



After



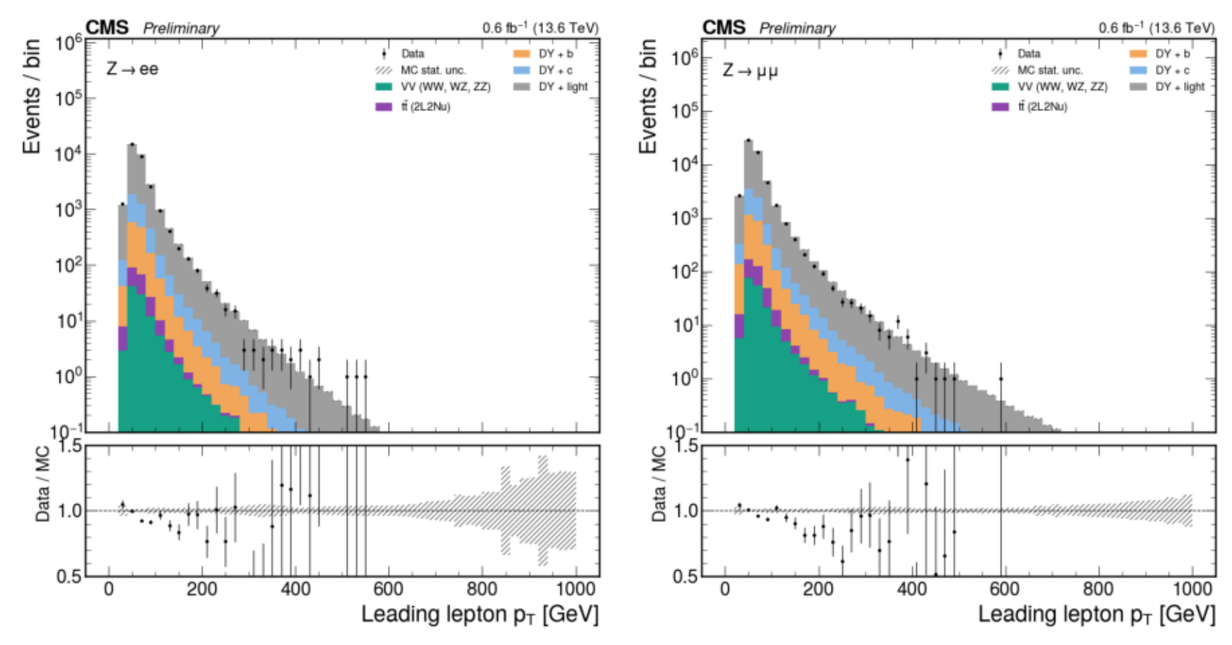
Level about 2% higher; shape slightly worse

Above 60 GeV MC is now 5–10% low: Z→μμ 1.00–1.07 → 1.05–1.10, Z→ee 0.99–1.02 → 1.05–1.07. Below 60 GeV still 0.94–0.97.

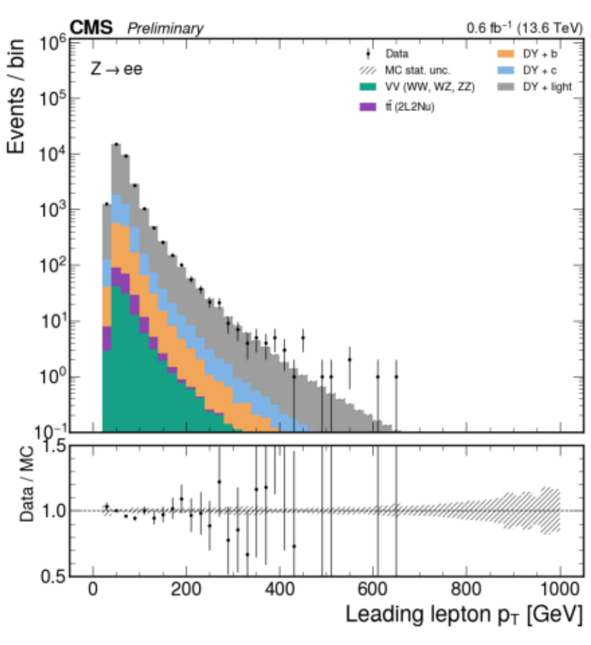
Leading-lepton pT, 2023 C: before and after

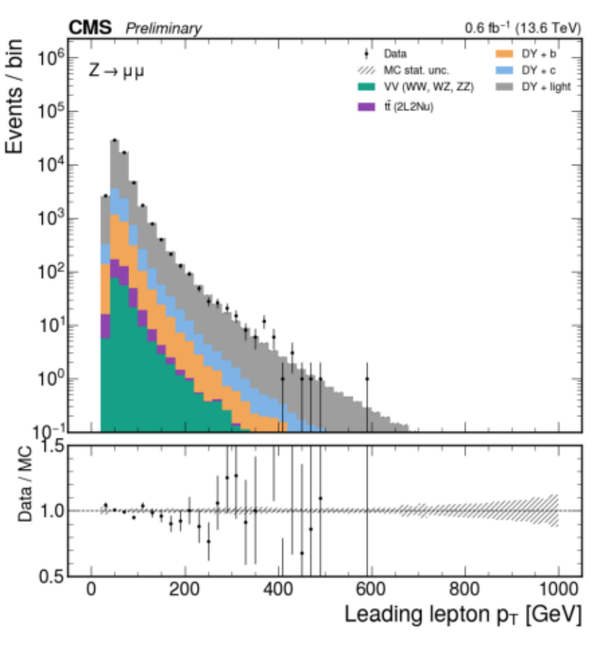
Left: before. Right: after (electrons' jets out of the MET + DY Z pT weights). Same files.

Before



After





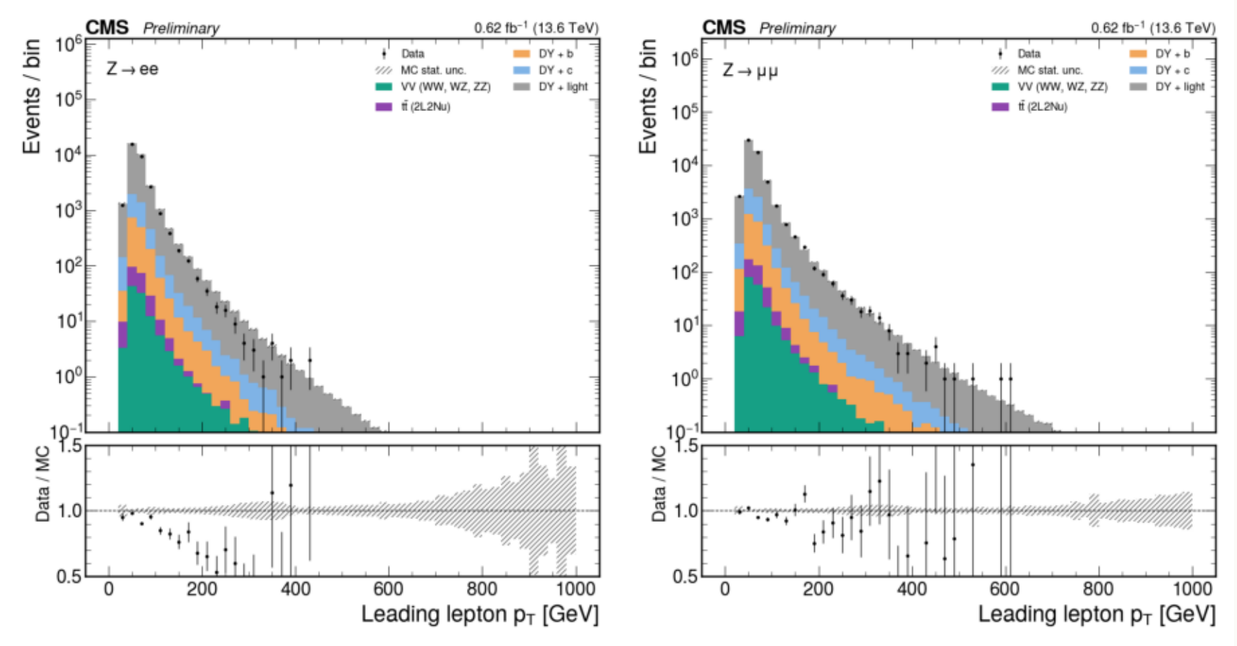
Better at high pT

Z→ee 140–300 GeV: 0.82–0.90 → 0.97–1.01. Z→μμ 140–300 GeV: 0.81–0.86 → 0.94–0.96. Each bin moves by 1–2σ.

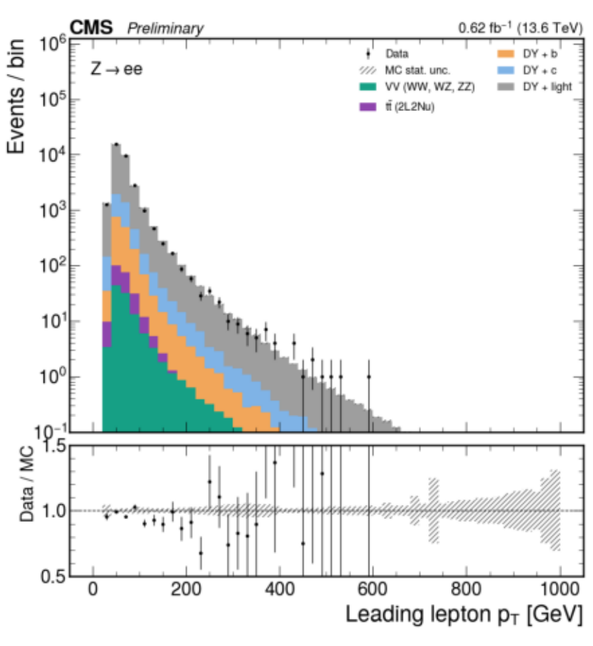
Leading-lepton pT, 2023 D: before and after

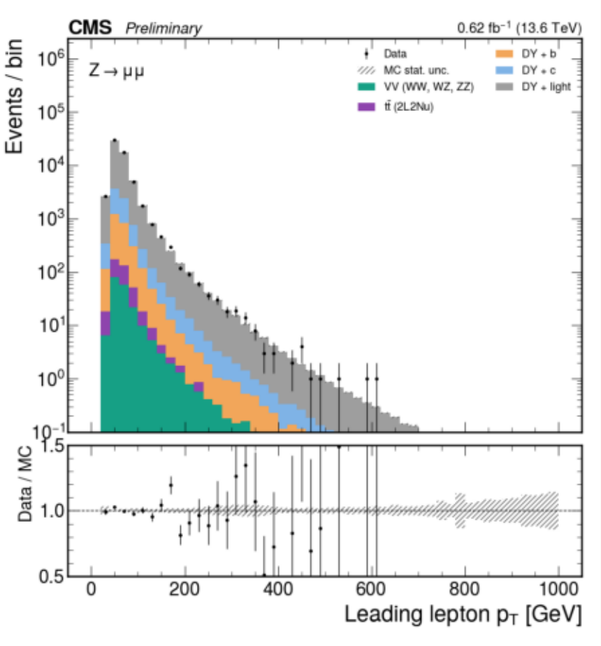
Left: before. Right: after (electrons' jets out of the MET + DY Z pT weights). Same files.

Before



After





The Z→ee drop at high pT is gone

Z→ee 200–300 GeV: 0.61 → 0.91, above 300: 0.53 → 1.03. Z→μμ barely changes. Electrons are still about 3% below muons overall.